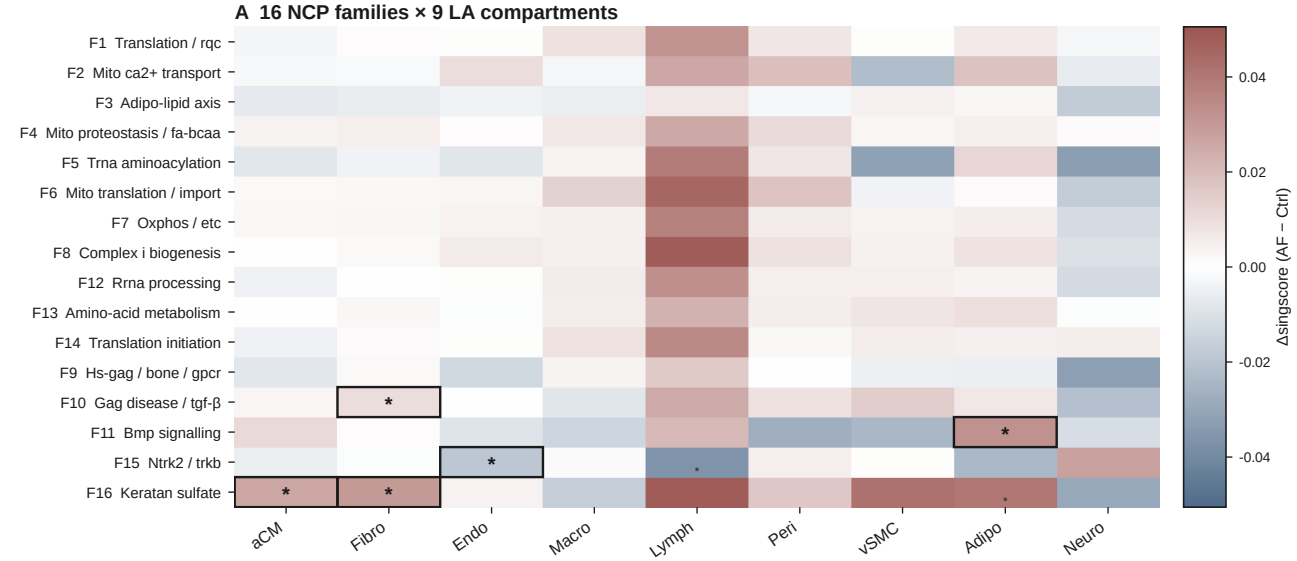
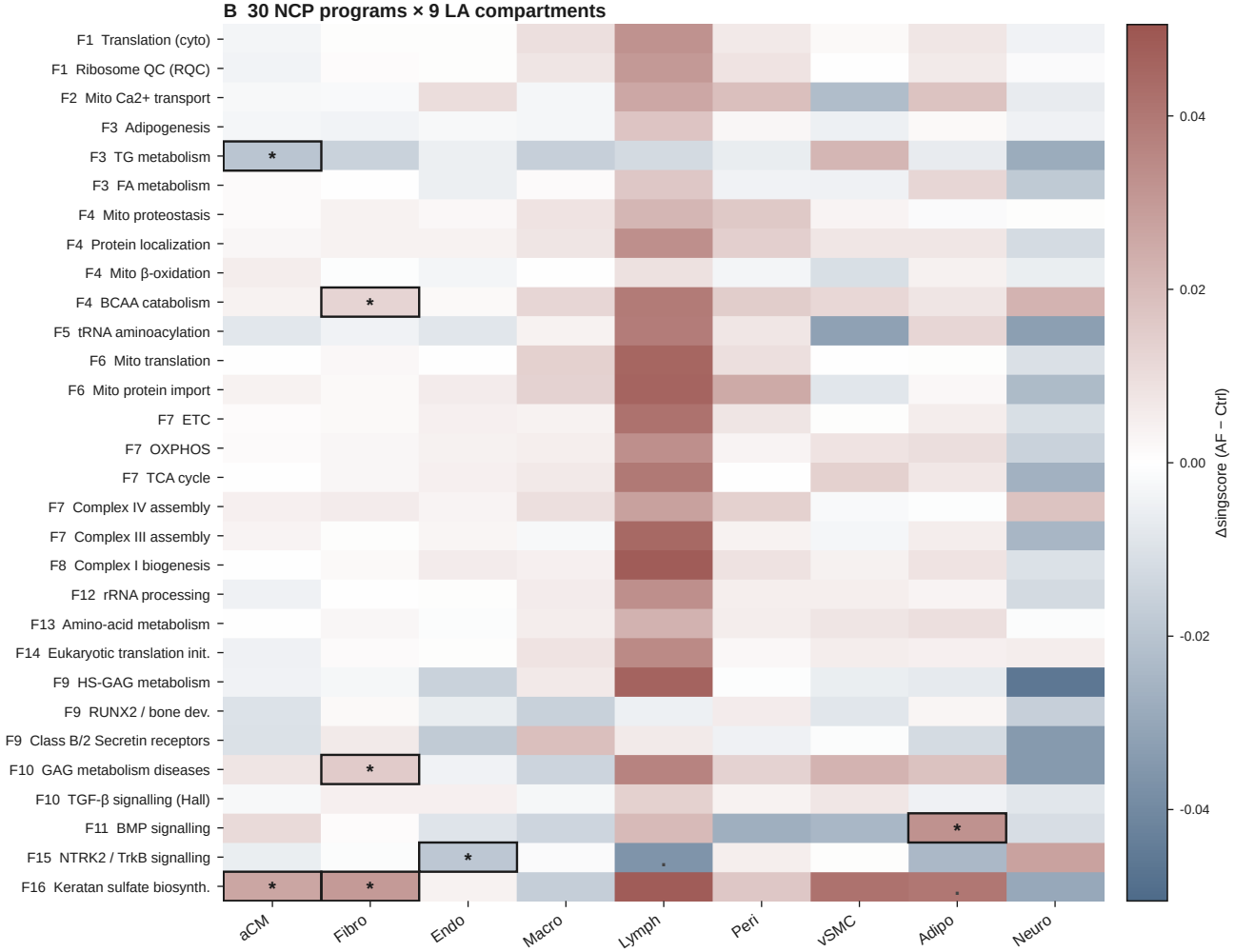
Full LA single-nucleus family and program localization

NCP · GSE255612 LA · exact-reference singscore lineage · Fidelity F10 (fibroblast); Relay F11 (adipocyte), F15 (endothelial only); Extension F16 (fibroblast + atrialCM); Below-detection 12 · F4 remains Below-detection (no fifth non-Below class) · nominal p (boxed * <0.05; dot <0.10); not FDR discovery · LA-only

A 16 NCP families × 9 LA compartments



B 30 NCP programs × 9 LA compartments



C Carrier-relative classes (nominal permutation; LA-only; not FDR-controlled)

Family	Class	Supported compartment(s)
F1	Below-detection	—
F2	Below-detection	—
F3	Below-detection	—
F4	Below-detection	—
F5	Below-detection	—
F6	Below-detection	—
F7	Below-detection	—
F8	Below-detection	—
F12	Below-detection	—
F13	Below-detection	—
F14	Below-detection	—
F9	Below-detection	—
F10	Fidelity	fibroblast
F11	Relay	adipocyte
F15	Relay	endothelial
F16	Extension	atrialCM,fibroblast